✅ 1. COMPLETE JAVA DEVELOPER ROADMAP

A. Computer Science Basics

- OS: process, threads, deadlock, memory mgmt.
- Networking: HTTP/HTTPS, DNS, TCP/UDP
- DBMS: ACID, normalization, indexing
- DSA basics: arrays, strings, hash map, recursion

B. Core Java

1. Basics

- Data types, variables, operators
- Loops, conditions
- String, StringBuilder, String Buffer
- Arrays, Wrapper classes

2. OOP (VERY IMPORTANT)

- Encapsulation
- Inheritance
- Polymorphism
- Abstraction
- Interfaces vs Abstract classes
- Overloading vs Overriding

3. Java 8 Features

- Streams
- Lambdas
- Functional interface
- Optional
- Method reference

4. Collections

- List, Set, Map
- ArrayList, LinkedList
- HashMap, LinkedHashMap, TreeMap
- HashSet, TreeSet

- Queue, Priority Queue
- Iterator, Comparator

5. JVM Internals

- JVM, JDK, JRE
- Memory areas
- GC working
- Class loader

C. SQL & Databases

SQL

- SELECT, INSERT, UPDATE, DELETE
- WHERE, LIKE, IN, BETWEEN
- JOINS (inner, left, right)
- GROUP BY & HAVING
- Indexes
- Constraints
- Functions: COUNT, SUM, MIN, MAX

RDBMS Tools

- MySQL
- PostgreSQL

NoSQL Basics

- MongoDB documents
- Aggregation pipeline

D. Spring & Backend Development

Spring Core

- IOC
- DI
- Beans
- Bean Scopes

Spring Boot

- Annotations
- Folder structure
- application. properties
- Embedded server
- Actuator

REST APIs

- GET, POST, PUT, DELETE
- Status codes
- RequestBody
- PathVariable
- Exception Handling
- Validation

JPA + Hibernate

- Entity
- @Id, @GeneratedValue
- OneToOne, OneToMany, ManyToMany
- JPQL
- HQL
- Lazy vs Eager

E. Tools

- Git
- GitHub
- Maven / Gradle
- Postman
- Swagger
- Lombok
- Logback

F. Testing

- JUnit 5

- Mockito
- MockMvc
- Integration Testing

G. Microservices

- REST communication
- API Gateway
- Eureka
- Config Server
- Feign client
- Circuit Breaker (Resilience4j)

H. DevOps Basics

- Docker
- Dockerfile
- Image, Container
- CI/CD basics
- GitHub Actions / Jenkins

I. Cloud Basics

- AWS EC2
- S3
- RDS
- IAM

J. System Design (Beginner Level)

- Client-server
- Load balancer
- Cache
- Database sharding
- Queues (Kafka basics)

✅ 2. 6-MONTH WEEKLY STUDY PLAN (BEGINNER → JOB READY)

📅 Month 1 – Core Java

Week 1: Basics, loops, arrays
Week 2: OOPs
Week 3: Strings, Exception Handling
Week 4: Collections

📅 Month 2 – Advanced Java + DSA

Week 5: Java 8 features
Week 6: Multithreading
Week 7: DSA – arrays, strings, hashing
Week 8: DSA – recursion, two pointers, sliding window

📅 Month 3 – SQL + Spring Boot Basics

Week 9: SQL basics
Week 10: Joins, subqueries, normalization
Week 11: Spring Core + Spring Boot
Week 12: REST APIs + Validation

📅 Month 4 – JPA + Project Work

Week 13: JPA/Hibernate mapping
Week 14: Services, DTO, Model Mapper
Week 15: Authentication (JWT)
Week 16: Build Project 1 (CRUD Project + SQL)

📅 Month 5 – Microservices + DevOps

Week 17: Microservices
Week 18: API Gateway + Eureka
Week 19: Docker
Week 20: Create Microservice project

📅 Month 6 – Interview Preparation

Week 21: 300 Java questions
Week 22: SQL + Spring Boot interview
Week 23: System design basics
Week 24: Resume + GitHub polish + Mock Interviews

✅ **3. GitHub Folder Structure (Professional MNC Standards)**

```
📦 java-developer-portfolio
 ├ 📁 core-java-practice
 ├ 📁 dsa-problems
 ├ 📁 springboot-crud-project
 │  ├ 📁 src
 │  ├ 📄 README.md
 ├ 📁 microservices-project
 │  ├ 📁 user-service
 │  ├ 📁 order-service
 │  ├ 📁 discovery-server
 │  ├ 📁 api-gateway
 │  ├ 📄 README.md
 ├ 📁 sql-practice
 │  └ 📄 queries.sql
 ├ 📄 resume.pdf
 └ 📄 profile-summary.md
```

✅ ****4. 300+ Interview Questions**

(For Java, Spring Boot, SQL, Microservices, Cloud, System Design)**

Core Java (60+)

- Why is String immutable?
- Difference between HashMap & HashTable
- What is overriding?
- What is functional interface?
- How does HashMap work internally?
- What is garbage collector?
- How many types of class loaders?

OOPs (30+)

- Explain OOP pillars with examples
- Interface vs abstract class
- Composition vs Aggregation

Java 8 (20+)

- Stream API internal working
- difference between map() and flatMap()

Collections (30+)

- How HashSet works internally?
- Difference between List & Set

SQL (30+)

- Write query to get 2nd highest salary
- What is indexing?
- Difference between INNER JOIN & LEFT JOIN

Spring Boot (40+)

- What is dependency injection?
- @Component vs @Service vs @Repository
- What is @ControllerAdvice?
- Spring bean scopes
- Autowiring types

JPA/Hibernate (30+)

- What is Persistence Context?
- Dirty Checking
- N+1 problem

Microservices (25+)

- What is API gateway?
- Service discovery
- Circuit breaker

System Design (20+)

- What is load balancer?
- Cache vs DB
- Horizontal vs Vertical scaling

Cloud & DevOps (15+)

- What is Docker?
- Container vs VM
- AWS EC2 vs S3

TOTAL = 300+ Questions

✅ **5. Projects You MUST Build (With Features + Tech Stack)**

📌 **Project 1: Employee Management CRUD**

- Java
- Spring Boot
- MySQL
- Postman
- JPA
- Lombok

Features:

- Add employee
- Update
- Delete
- Get by id
- Get all
- Validation + Exception Handling

📌 **Project 2: E-Commerce Backend**

Tech: Spring Boot, MySQL, JWT, JPA, ModelMapper

Features:

- Signup/Login (JWT)
- Product API
- Category API
- Cart API
- Order API
- Roles (ADMIN/USER)

📌 **Project 3: Microservices Project**

Tech: Spring Cloud, Eureka, Gateway, Feign, MySQL

Services:

- User Service
- Order Service
- Product Service
- API Gateway
- Discovery Server

✅ **6. Daily Routine for Becoming Java Developer (BEST)**

⏰ **6 Hours Daily Plan**

- 1 hr → Java
- 1 hr → DSA
- 1 hr → SQL
- 1.5 hr → Spring Boot
- 30 min → Git, DevOps
- 1 hr → Project building